Candidate proofs for the irrationality exponents of $\log 2$ and rational logarithms

Research draft prepared with Codex

8 October 2026

Abstract

This document gives the full proposed interpolation and determinant argument for the irrationality exponent of the natural logarithm of two. It adapts the separated-weight method in OpenAI's manuscript on π, replacing centers with $Y = 1$ by distinct multiplicative centers $Y = 2^j$. A separate section extends the argument to $\log a$ for positive rational $a \neq 1$, nonzero rational linear combinations of such logarithms, and positive algebraic numbers with a positive integral power in $\mathbb{Q}$. The crucial generalized interpolation statement and the resulting proof are candidates requiring further verification. This document does not establish a certified new theorem.

Verification status. The new distinct-Y geometric interpolation argument has been read internally, but has not been independently certified or formalized. Parameter calculations and successful TeX compilation do not validate that argument. Every proof below is presented as part of the candidate derivation, with its exact claim available for scrutiny.

Contents

1 Claim, provenance, and verification boundary **2**

2 Interpolation with separated weights **2**
2.1 A local multiplicity comparison . . . 3
2.2 The curve inequality . . . 5
2.3 From the curve inequality to all coefficient packets . . . 7

3 The arithmetic and analytic determinant estimates **9**
3.1 A full-row-rank minor and its arithmetic size . . . 10
3.2 Translation to entire functions . . . 11
3.3 Repeated Taylor degrees force a quadratic saving . . . 13
3.4 A comparison with two alternatives . . . 14

4 Choice of parameters and proof of Theorem 1.1 **15**

5 Extension to positive rational logarithms **17**
5.1 Exact arithmetic clearing for a rational base . . . 18
5.2 Translation and analytic collision with either sign . . . 19
5.3 Finite parameter choice for each fixed base . . . 20
5.4 Rational scaling and further conditional consequences . . . 21

6 Audit interface for the candidate proof **22**

1 Claim, provenance, and verification boundary

For an irrational real number x, use

$$\mu(x) = \sup\{\nu > 0 : 0 < |x - p/q| < q^{-\nu} \text{ for infinitely many coprime } p \in \mathbb{Z},\ q \geq 2\}.$$

Dirichlet's theorem gives $\mu(x) \geq 2$. The published bound used here as background is Marcovecchio's $\mu(\log 2) < 3.57455391$ [2, Theorem 1.1].

Sections 2–4 adapt OpenAI's *The irrationality exponent of π is 2*, dated 24 September 2026 [1]. Credit for the local multiplicity, projective interpolation, determinant and parameter mechanisms belongs to that manuscript and the earlier work cited below. The original source is publicly linked in the references; the Apache-2.0 license and adaptation notice accompany this manuscript. This standalone source has no external TeX inputs, figures, or bibliography files.

The substantive proposed modification is to use centers $(2^j, jp_1/q_1, \ldots, jp_m/q_m)$ and local coordinates

$$Y = 2^j(1 + t), \qquad X_i = jp_i/q_i + u_i + \log(1 + t).$$

The new geometric statement must allow successive weight thresholds to be selected independently of the center coordinates. Only the eventual degree threshold may depend on the fixed centers. This order of quantifiers is essential to the final contradiction.

The posted Lean implementation for the source π theorem has not been rebuilt or exhaustively audited here. It does not certify this proposed $\log 2$ extension. No exhaustive novelty claim is made. An external mathematical audit or an appropriate formal proof remains necessary to establish the candidate result.

Theorem 1.1 (Proposed result; candidate proof)**.** *For every real $\nu > 2$, there is an integer $Q(\nu)$ such that*

$$|\log 2 - p/q| \geq q^{-\nu} \qquad (p \in \mathbb{Z},\ q \in \mathbb{Z},\ q \geq Q(\nu)).$$

Consequently $\mu(\log 2) = 2$.

The remaining sections give the proposed argument, rather than a claim that its correctness has already been established.

2 Interpolation with separated weights

We prove the interpolation statement used in the determinant argument. Its essential uniformity is that the weights can be chosen before the interpolation points: each coordinate weight has a lower threshold depending on the earlier weights, but not on the locations of the points. The degree at which interpolation becomes surjective is allowed to depend on all the fixed data.

All multi-indices below have nonnegative integer entries. For a positive rational vector $W = (w_0, \ldots, w_m)$, the W-degree of the monomial $Y^h X_1^{\alpha_1} \cdots X_m^{\alpha_m}$ is $w_0 h + \sum_i w_i \alpha_i$. Write $\mathcal{P}_W(H)$ for the complex vector space of polynomials of W-degree at most H. For a positive rational vector $V = (v_0, \ldots, v_m)$, let $\mathcal{J}_V(H)$ be the quotient of $\mathbb{C}[[t, u_1, \ldots, u_m]]$ by the ideal of series whose monomials all have V-weight at least H. Thus a vector in $\mathcal{J}_V(H)$ is a packet of coefficients indexed by

$$v_0 s + \sum_{i=1}^{m} v_i \beta_i < H.$$

The strict inequality in this definition will be used throughout.

Theorem 2.1 (Separated-weight interpolation)**.** *Fix integers $m, K \geq 1$ and positive rational numbers w_0, v_0, θ such that $0 < \theta < 1$ and*

$$K\theta^m < 1, \qquad K\frac{w_0}{v_0}\theta^m < 1. \tag{2.1}$$

There are successive lower thresholds for positive rational numbers $w_1, \dots, w_m$, depending only on these fixed data and the previously chosen weights, with the following property. Put

$$W = (w_0, w_1, \dots, w_m), \qquad V = (v_0, w_1/\theta, \dots, w_m/\theta).$$

For any points

$$a_j = (y_j, c_{j1}, \dots, c_{jm}) \in \mathbb{C}^{m+1}, \qquad 0 \leq j < K,$$

where $y_0, \dots, y_{K-1}$ are nonzero and pairwise distinct, the map

$$\begin{aligned} \mathcal{P}_W(H) &\longrightarrow \bigoplus_{j=0}^{K-1} \mathcal{J}_V(H), \\ P &\longmapsto \big(P(y_j(1+t), (c_{ji} + u_i + \log(1+t))_{i=1}^m)\big)_{j=0}^{K-1}. \end{aligned} \tag{2.2}$$

is surjective for all sufficiently large, sufficiently divisible integers H. The divisibility requirement can be chosen using only the weights. The remaining lower threshold for H may depend on the points.

We first establish a numerical inequality on algebraic curves, and then use it to obtain all the jets in (2.2). The local estimate driving the curve argument is recorded separately.

2.1 A local multiplicity comparison

A collection of vector fields is called a *normal basis* to a smooth subvariety at a point if its images form a basis of the ambient tangent space modulo the tangent space of the subvariety. For an irreducible subvariety, this terminology will refer to a general smooth point. There are only finitely many possible index sets in the applications below, so all their generic rank conditions can be checked on one common open subset.

Lemma 2.2 (Weighted transverse multiplicity)**.** *Give the coordinates $x_1, \dots, x_d$ positive rational degree weights $\rho_1, \dots, \rho_d$. On an open subset of $\mathbb{C}^d$, let $D_1, \dots, D_d$ be commuting analytic vector fields forming a frame, and assign them positive rational costs $\kappa_1, \dots, \kappa_d$. Fix $\varepsilon > 0$.*

Let $N > 0$. Suppose that polynomials $f_1, \dots, f_r$ have weighted degree at most N, that Z is an irreducible component of their common zero set of codimension $k \geq 1$, and that

$$D^\gamma f_\ell\big|_Z = 0 \quad \text{whenever} \quad \sum_b \kappa_b \gamma_b < \varepsilon N. \tag{2.3}$$

For any coordinate normal basis A and any D-field normal basis B, each of cardinality k, one has

$$\prod_{a \in A} \rho_a \leq 2k!\, \varepsilon^{-k} \prod_{b \in B} \kappa_b \tag{2.4}$$

The bound is uniform in the equations, the component Z, and the analytic frame; no lower threshold for N is required.

Proof. Choose a smooth point x of Z where both normal-basis conditions hold and which lies on no other irreducible component of the common zero set. Flowing from Z along the fields indexed by B gives local analytic coordinates (z, b), where z parametrizes Z and $b = (b_1, \dots, b_k)$ consists of transverse flow parameters. Commutativity implies that the chosen fields are the derivatives in the b-directions. Thus (2.3) says that in the expansion

$$f_\ell(z, b) = \sum_{\alpha \in \mathbb{Z}^k_{\geq 0}} f_{\ell,\alpha}(z) b^\alpha,$$

every coefficient function of B-weight less than εN vanishes identically on the chosen neighborhood in Z.

Let S be the affine coordinate slice through x obtained by fixing the coordinates outside A. It is transverse to Z, and the zero set of the restricted equations is isolated at x. The flow parameters b give regular analytic coordinates on S; the coordinates z become analytic functions $z(b)$. Substitution cannot generate a missing term of smaller B-weight. Indeed, each surviving summand already contains a monomial b^α of weight at least εN, and its analytic coefficient has only nonnegative powers of b. Consequently the ideal of the restricted equations is contained in the monomial ideal of weight at least εN. Its quotient therefore has length at least

$$\# \left\{ \alpha \in \mathbb{Z}^k_{\geq 0} : \sum_{b \in B} \kappa_b \alpha_b < \varepsilon N \right\} \geq \frac{(\varepsilon N)^k}{2k! \prod_{b \in B} \kappa_b} \tag{2.5}$$

In fact the factor $1/2$ can be omitted: the simplex $\{x \in \mathbb{R}^k_{\geq 0} : \sum_{b \in B} \kappa_b x_b < \varepsilon N\}$ is covered by the disjoint half-open unit cubes $\alpha + [0, 1)^k$ indexed by the admissible multi-indices $\alpha = \lfloor x \rfloor$. Its volume is $(\varepsilon N)^k / (k! \prod_{b \in B} \kappa_b)$. Thus the displayed bound holds for every $N > 0$. Analytic local coordinates preserve finite length, so no radius or coefficient bound for the flow map is needed.

In the regular local ring of the k-dimensional slice, the ideal generated by the restricted equations is primary to the maximal ideal. Over the infinite field $\mathbb{C}$, k sufficiently general constant linear combinations of these equations still have an isolated common zero at x: successive generic combinations avoid the finitely many positive-dimensional components that remain. Their ideal is contained in the original ideal. Their quotient length is therefore an upper bound for the length in (2.5).

Each of the k combinations has weighted degree at most N in the variables indexed by A. We claim that their local intersection length is at most

$$\frac{N^k}{\prod_{a \in A} \rho_a}. \tag{2.6}$$

Choose an integer M_0 clearing the denominators of the ρ_a, and make the substitutions

$$x_a = \lambda_a + z_a^{M_0 \rho_a} \qquad (a \in A).$$

Choose the constants λ_a so that no preimage of the given point is ramified. There are $\prod_{a \in A} (M_0 \rho_a)$ such preimages, each with the same local intersection length. The substituted polynomials have ordinary total degree at most $M_0 N$. The ordinary Bézout bound for isolated complete-intersection zeros, with their local multiplicities, bounds the sum of these lengths by $(M_0 N)^k$; see [3, Corollary VIII.3, p. 144]. The multiplicities there are local quotient lengths, unchanged by completion [3, Section IV.4.2]. Dividing gives (2.6). This application concerns isolated zeros only and permits other components elsewhere.

Combining (2.5) and (2.6) proves the result. Neither the polynomials nor the possibly varying component enters the constant in (2.4). □

2.2 The curve inequality

Choose a positive rational number σ sufficiently small that

$$\begin{aligned}(1+3\sigma)^{m+1}K(w_0/v_0)\theta^m &< 1,\\ (1+3\sigma)^m K\theta^m &< 1,\\ (1+\sigma)\theta &< 1.\end{aligned} \tag{2.7}$$

This choice is possible by (2.1). Set $v_i = w_i/\theta$ for $i > 0$. Fix $L_0 = m+2$ and put

$$C_* = \max_{1\le k\le m} 2k!(L_0/\sigma)^k. \tag{2.8}$$

We choose the positive weights successively so that for every pair of subsets $A, B \subseteq \{0,\dots,m\}$ of the same cardinality between 1 and m,

$$\prod_{a\in A} w_a > C_* \prod_{b\in B} v_b \tag{2.9}$$

whenever the largest positive index in their symmetric difference belongs to A.

Here and below a condition involving that largest index is imposed only when a positive index does differ. The successive choice is possible: if this largest index is i, the ratio in (2.9) is w_i times a positive expression in the already chosen lower weights and the fixed zeroth weights. Every higher index occurs in both sets or in neither; in the former case its ratio is $w_j/v_j = \theta$. Hence the expression is independent of all higher weights. There are finitely many pairs of sets. A sufficiently large choice of w_i enforces all conditions whose largest differing index is i, and subsequent choices do not disturb them. This construction involves no interpolation point.

Let C be an irreducible algebraic curve whose generic point lies in the affine coordinate chart. Such a curve has a smooth projective model over $\mathbb{C}$; see [4, Theorem 53.2.6 and Lemma 53.2.8, Tag 0BXX]. On that model define

$$\deg_W C = \sum_P \max\left\{0, -\frac{\operatorname{ord}_P Y}{w_0}, -\frac{\operatorname{ord}_P X_1}{w_1}, \dots, -\frac{\operatorname{ord}_P X_m}{w_m}\right\}. \tag{2.10}$$

The order of an identically zero coordinate is understood as $+\infty$; it contributes no pole. At a branch P through a_j, set

$$h_P = \min\left\{\frac{\operatorname{ord}_P(Y-y_j)}{v_0}, \frac{\operatorname{ord}_P(X_1 - c_{j1} - \log(Y/y_j))}{v_1}, \dots, \frac{\operatorname{ord}_P(X_m - c_{jm} - \log(Y/y_j))}{v_m}\right\}. \tag{2.11}$$

The logarithm here is its formal expansion at $Y = y_j$, namely $\log(Y/y_j)$. All entries have positive order, possibly infinite, and their minimum is finite on a nonconstant curve.

Proposition 2.3 (Curve inequality)**.** *With the weights chosen above, every such curve satisfies*

$$\deg_W C \ge (1+\sigma)\sum_{P\mapsto a_j} h_P, \tag{2.12}$$

where the sum includes all branches through the prescribed points.

Proof. There is nothing to prove if the curve misses all the points. Suppose instead that (2.12) fails. For every sufficiently large auxiliary integer N, linear algebra gives a nonzero polynomial F_N

of W-degree at most N whose expansions at all the points have V-order at least $(1+3\sigma)N$. Indeed, the number of its coefficients has leading term

$$\frac{N^{m+1}}{(m+1)!\prod_{a=0}^{m} w_a},$$

whereas the number of conditions has leading term

$$\frac{K(1+3\sigma)^{m+1}N^{m+1}}{(m+1)!\prod_{a=0}^{m} v_a}.$$

Their comparison is the first inequality in (2.7). This argument requires no independence among the conditions.

On the open set $Y \neq 0$ consider the commuting frame

$$D_0 = Y\partial_Y + \sum_{i=1}^{m} \partial_{X_i}, \qquad D_i = \partial_{X_i} \quad (1 \le i \le m). \tag{2.13}$$

It preserves polynomial degree at most N. In the formal coordinates of (2.2),

$$D_0 = (1+t)\partial_t, \qquad D_i = \partial_{u_i}.$$

Thus differentiation by D_a decreases weighted order by at most v_a. Every $D^\gamma F_N$ with $\sum_a v_a\gamma_a \le \sigma N$ retains order at least $(1+2\sigma)N$ at each point. On a branch P, its order is consequently at least $(1+2\sigma)Nh_P$.

A polynomial of W-degree at most N restricts to a rational function on the complete model of C with total pole degree at most $N\deg_W C$. Since we assumed that (2.12) fails, the just obtained total zero order exceeds this pole bound. The ordinary divisor equality for a nonzero rational function therefore forces

$$D^\gamma F_N\big|_C = 0 \qquad \left(\sum_a v_a\gamma_a \le \sigma N\right). \tag{2.14}$$

All the assertions about branch orders hold for formal power series: each monomial has order at least its weight times h_P, and cancellation can only increase the order.

We have converted excess contact into vanishing of many derivatives on C. We next find a component on which that vanishing persists, as in the derivative-ideal method of zero estimates [7, Section 5], and compare all its normal bases. Put $\delta = \sigma N/L_0$. For $0 \le r \le L_0$, let V_r be the common zero set in $Y \neq 0$ of the derivatives of cost at most $r\delta$. Let C° be the part of C in this affine open set. The V_r form a decreasing chain, and (2.14) puts C° in V_{L_0}. Choose an irreducible component of V_{L_0} containing C°, and extend it backwards to nested irreducible components of the earlier V_r. Every selected component has dimension between 1 and m: it contains a curve, and it lies in the proper zero set of F_N. Since there are $L_0+1 = m+3$ levels, two adjacent selected components coincide. Denote this persistent component by Z, and write $k = \operatorname{codim} Z$.

Take the derivatives of cost at most $r\delta$ as the equations of the earlier of these two levels. Any further derivative of cost less than δ vanishes along Z, because the fields commute and Z is also a component at the next level. Lemma 2.2, with $\varepsilon = \sigma/L_0$, gives

$$\prod_{a\in A} w_a \le C_* \prod_{b\in B} v_b \tag{2.15}$$

for every coordinate normal basis A and every frame normal basis B. The equations and Z may vary with N; the uniformity proved in Lemma 2.2 is exactly what permits this use.

We show that Y is constant on Z, working on the common smooth open subset where all the normal-basis conditions hold. Suppose first that some positive coordinate direction is nontangent to Z, and choose the largest such index i. Every larger positive index occurs in neither kind of normal basis, since $D_j = \partial_{X_j}$ for $j > 0$. Some coordinate normal basis A contains i. If a frame normal basis B omitted i, the largest positive index where these sets differ would be i. Then (2.9) would contradict (2.15). Thus every frame normal basis contains D_i.

The other frame vectors consequently fail to span the normal quotient. Their ambient span is the hyperplane

$$\ker(dX_i - dY/Y).$$

For an ambient hyperplane Q and a tangent space T, the image of Q in the quotient by T fails to be surjective precisely when $T \subseteq Q$. We obtain

$$dX_i = dY/Y \quad \text{on } Z. \tag{2.16}$$

This identity forces Y to be constant on the algebraic variety Z. Indeed, if Y were nonconstant, choose a smooth point where $dY \neq 0$ and intersect with general algebraic hyperplanes through that point until a curve remains on which Y is nonconstant. On the smooth complete model of this curve, (2.16) is an identity of meromorphic differentials. A nonconstant meromorphic function Y has a zero or pole, where dY/Y has nonzero residue $\operatorname{ord}_P Y$. An exact differential dX_i has zero residue at every point. This is a contradiction. It follows also that X_i is constant, since $dX_i = 0$ in characteristic zero.

If instead every positive coordinate direction is tangent, the m independent vectors ∂_{X_i} all lie in the tangent space of Z. Properness forces that tangent space to be their span, and $dY = 0$ on Z again. We conclude in all cases that Y is constant on Z, and hence on C. Since the nonzero numbers y_j are pairwise distinct, C passes through at most one prescribed center, say a_j. At least one X_l is nonconstant on C, since C is a curve and Y is constant. On every branch through this center, $Y = y_j$ and therefore $\log(Y/y_j) = 0$. The nonzero rational function $X_l - c_{jl}$ has total pole degree at most $w_l \deg_W C$, whereas its order at each of these branches is at least $(w_l/\theta)h_P$. Consequently

$$\deg_W C \geq \theta^{-1} \sum_{P \mapsto a_j} h_P > (1+\sigma) \sum_{P \mapsto a_j} h_P.$$

This contradicts the assumed failure of the curve inequality. The original proof's second argument inside the fiber $Y = 1$ is unnecessary here. The hypothesis $K\theta^m < 1$ has been retained to keep the subsequent parameter choice identical to the source proof. □

2.3 From the curve inequality to all coefficient packets

We complete the proof of Theorem 2.1. The argument uses ordinary projective schemes and the ordinary blow-up; smoothness or normality of the compactification will not be needed. Curve positivity and asymptotic jet generation are closely related in the Seshadri-constant approach of [6, Section 6]. Here we supply the passage for our weighted ideals at several points, including the eventual ordinary-power comparison required on a possibly singular scheme.

Choose an integer R sufficiently large that R/w_a and R/v_a are positive integers for every index. Embed affine space by all monomials of W-degree at most R, including the constant monomial, and let X be the projective closure of the image. Since these monomials include each coordinate, the chart of the constant monomial is precisely the original affine space. Let $L = \mathcal{O}_X(1)$, a very ample line bundle.

For any curve meeting this chart,

$$\deg(L|_C) = R \deg_W C. \tag{2.17}$$

To see this on its smooth complete model, at each point clear the largest pole of the homogeneous monomial coordinates. Every monomial has pole order at most R times the maximum in (2.10). A pure power of one coordinate attains that bound whenever the maximum is positive, and the constant monomial attains it when the maximum is zero. The resulting sections have no common zero, proving (2.17).

To make the local data algebraic, replace $\log(1+t)$ in each coordinate by a fixed Taylor polynomial $G_i(t)$ whose omitted terms have t-weight strictly greater than v_i. Then

$$t = Y/y_j - 1, \qquad u_i' = X_i - c_{ji} - G_i(t)$$

are regular local coordinates at a_j. This replacement preserves h_P. Indeed, if t is not identically zero on a branch, each omitted term has order divided by v_i strictly larger than $\operatorname{ord}_P(t)/v_0$; if t is identically zero, its tail vanishes. It also preserves interpolation surjectivity for every threshold: the formal coordinate change fixes t and changes u_i only by terms of weight greater than v_i. It and its inverse preserve the weighted filtration and induce inverse triangular maps on each finite coefficient packet.

At a_j form the finite-colength ideal

$$I_j = \left(t^{R/v_0}, (u_1')^{R/v_1}, \dots, (u_m')^{R/v_m}\right).$$

The points are distinct. These local ideals, together with the unit ideal away from the points, define a coherent ideal sheaf I on X. For a branch through a center,

$$\operatorname{ord}_P I = \min_a \{(R/v_a) \operatorname{ord}_P z_a\} = R h_P, \qquad z_0 = t, \quad z_i = u_i'. \tag{2.18}$$

Let $p : X' \to X$ be the ordinary blow-up of I, namely the relative Proj of its Rees algebra, and write

$$I\mathcal{O}_{X'} = \mathcal{O}_{X'}(-E).$$

The scheme X is integral and Noetherian, I is a coherent nonzero ideal, and E is an effective Cartier divisor. The tautological bundle $\mathcal{O}_{X'}(-E)$ is relatively ample; these are the standard blow-up properties in [4, Tags 02OS, 02ND, and 02NS].

Write $A = p^*L$ in additive divisor notation. We claim that

$$A - (1+\sigma)E \quad \text{is nef.} \tag{2.19}$$

For a noncontracted integral curve Γ upstairs whose image meets the affine chart, the map to its image is birational, since p is an isomorphism away from the finite center. On the complete normalizations, (2.17) and (2.18) give

$$A \cdot \Gamma = R \deg_W p(\Gamma), \qquad E \cdot \Gamma = R \sum_{P \mapsto a_j} h_P.$$

Proposition 2.3 gives the desired nonnegative degree. A curve whose image misses the affine chart avoids the exceptional divisor. A contracted curve has A-degree zero and positive degree for $-E$ by relative ampleness. These cases account for every integral curve, proving (2.19).

For sufficiently large $a > 1$, the divisor $aA - E$ is ample. One can see this directly: choose b so that $I \otimes L^b$ is globally generated. The Rees construction embeds X' in a product $X \times \mathbb{P}^r$, with the second hyperplane bundle restricting to $bA - E$. Tensoring with the first very ample bundle makes $(b+1)A - E$ ample. Now the identity

$$A - E = \frac{a-1}{a(1+\sigma)-1}(A - (1+\sigma)E) + \frac{\sigma}{a(1+\sigma)-1}(aA - E) \tag{2.20}$$

expresses $A - E$ as a positive multiple of a nef class plus a positive multiple of an ample class. It is therefore ample; see [5, Corollary 1.4.10]. This theorem applies to projective schemes and does not require a smooth blow-up.

For all sufficiently large n, the graded-piece and relative vanishing theorems for the ordinary Rees algebra give

$$p_*\mathcal{O}_{X'}(-nE) = I^n, \qquad R^q p_*\mathcal{O}_{X'}(-nE) = 0 \quad (q > 0). \tag{2.21}$$

Indeed, on any affine open of the Noetherian base the Rees algebra is a Noetherian graded algebra generated in degree one. Its sufficiently high graded pieces recover the sections of its associated sheaf on Proj, and the higher cohomology of those twists vanishes. A finite affine cover gives one common threshold; see [4, Tags 0AG6 and 0AG7]. In particular, (2.21) uses ordinary powers I^n, not their integral closures, and makes no assertion about small n.

By the projection formula, Leray, and Serre vanishing for the ample line bundle $A - E$,

$$H^1(X, I^n \otimes L^n) = H^1(X', \mathcal{O}_{X'}(n(A - E))) = 0 \tag{2.22}$$

for all sufficiently large n; see [4, Tag 0B5U]. The exact sequence for I^n consequently gives a surjection from $H^0(X, L^n)$ onto all its local coefficient data modulo I^n. The constant projective coordinate trivializes L on the affine chart. Finite-colength local quotients are unchanged by completion, so these algebraic classes are also the formal coefficient data used in the theorem. In these trivializations, every element of I^n has weighted order at least nR, because every displayed generator of I has weight R. Thus

$$I^n \subseteq \{\text{local series of weight at least } nR\}. \tag{2.23}$$

Surjectivity modulo I^n therefore implies surjectivity onto the smaller quotient consisting of the desired packets of weight less than nR. Equality in (2.23) is not needed.

Finally, every section of L^n for sufficiently large n is supplied by a homogeneous degree-n polynomial in the projective embedding coordinates. This follows from Serre vanishing applied to the homogeneous ideal sheaf of X in its projective space. Such a polynomial restricts on the affine chart to one of W-degree at most nR. Taking $H = nR$ and undoing the filtered Taylor-polynomial coordinate change proves the surjectivity in (2.2), and completes the proof of Theorem 2.1.

3 The arithmetic and analytic determinant estimates

Interpolation now supplies the algebraic rank needed for an interpolation determinant, in the sense of [8, Section 6]. We estimate one full-row minor from below arithmetically, and then translate its rows to exact logarithmic points to obtain an analytic upper bound.

Throughout this section all parameters except H are fixed. Write

$$w\alpha = \sum_{i=1}^{m} w_i\alpha_i, \qquad w_* = \min_{1\le i\le m} w_i, \qquad \omega = \log 2.$$

Let $\nu > 2$, and suppose that integers p_i and $q_i \geq 2$ satisfy

$$\left|\log 2 - \frac{p_i}{q_i}\right| \leq q_i^{-\nu}, \qquad w_i = \lceil \log q_i \rceil, \qquad r_i = \frac{p_i}{q_i} \quad (1 \leq i \leq m). \tag{3.1}$$

Assume that $p_i \neq 0$ and that $m, K, w_0, v_0, \theta, w_1, \ldots, w_m$ satisfy the hypotheses of Theorem 2.1, with centers $y_j = 2^j$ and $c_{ji} = jr_i$, $0 \leq j < K$. In particular w_i is a positive integer for $i > 0$, while w_0 and v_0 are positive rational numbers. Fix $F_0 > 2/\theta$, and put

$$T_i = \left\lceil \frac{F_0 w_i}{v_0} \right\rceil, \qquad G_i(t) = \sum_{1 \leq k < T_i} \frac{(-1)^{k+1} t^k}{k}. \tag{3.2}$$

Empty sums are zero. The first omitted term has weight $v_0 T_i \geq F_0 w_i > w_i/\theta$. Consequently the substitution $u_i \mapsto u_i + G_i(t) - \log(1+t)$ is an invertible substitution preserving the weighted filtration. It induces an invertible map on every finite space of coefficients of weight less than H. Thus replacing the logarithms in Theorem 2.1 by these G_i preserves surjectivity.

3.1 A full-row-rank minor and its arithmetic size

Use the monomials

$$P(Y, X) = Y^h X^\alpha, \qquad h \in \mathbb{Z}_{\geq 0}, \quad \alpha \in \mathbb{Z}_{\geq 0}^m, \qquad w_0 h + w\alpha \leq H$$

as columns, and the triples

$$\rho = (j, s, \beta), \qquad 0 \leq j < K, \quad s \in \mathbb{Z}_{\geq 0}, \quad \beta \in \mathbb{Z}_{\geq 0}^m, \qquad v_0 s + \frac{w\beta}{\theta} < H$$

as rows. The matrix entry is the coefficient of $t^s u^\beta$ in

$$P\big(2^j(1+t), jr_1 + G_1(t) + u_1, \ldots, jr_m + G_m(t) + u_m\big). \tag{3.3}$$

For all sufficiently large, sufficiently divisible H, choose a square minor $\Delta_H \neq 0$ using every row, as permitted by Theorem 2.1. Denote its size by $M = M_H$, and set

$$\bar{b} = \bar{b}_H = \frac{1}{MH} \sum_{\rho = (j,s,\beta)} w\beta. \tag{3.4}$$

The elementary lattice count for a simplex gives

$$M \sim \frac{K \theta^m H^{m+1}}{(m+1)!\, v_0 \prod_{i=1}^m w_i}, \qquad 0 \leq \bar{b} \leq \theta. \tag{3.5}$$

Indeed, after scaling by H, the lattice boxes of side lengths H^{-1} are Riemann sums for the fixed simplex. Its volume is $\theta^m/((m+1)! v_0 \prod_i w_i)$; its boundary has volume zero, so the strict inequality in the row definition does not affect the leading term. We always let H tend to infinity only after all the weights and centers have been fixed.

Lemma 3.1 (Arithmetic lower bound)**.** *Let $\Lambda = 4 \log 2$, and define*

$$E_{\mathrm{ar}} = \frac{\Lambda F_0 m}{v_0} + \Lambda \sum_{i=1}^m \frac{1}{w_i} + \frac{\theta}{w_*}. \tag{3.6}$$

Then every minor just chosen satisfies

$$\frac{\log |\Delta_H|}{MH} \geq -(1 - \bar{b}) - E_{\mathrm{ar}}. \tag{3.7}$$

Proof. Let $L_i = \operatorname{lcm}(1, \dots, T_i - 1)$, interpreted as 1 if $T_i = 1$, and put

$$D_H = \prod_{i=1}^{m} L_i^{\lfloor H/w_i \rfloor}.$$

Scale a column $Y^h X^\alpha$ by $\prod_i q_i^{\alpha_i}$ and a row (j, s, β) by $\prod_i q_i^{-\beta_i}$. The scaled entry is zero unless $\alpha \geq \beta$ componentwise; otherwise it is

$$\prod_i \binom{\alpha_i}{\beta_i} 2^{jh} [t^s](1+t)^h \prod_i (q_i(jr_i + G_i(t)))^{\alpha_i - \beta_i}.$$

Here $q_i j r_i = j p_i \in \mathbb{Z}$. Since G_i has coefficients in $L_i^{-1}\mathbb{Z}$, the denominator of this entry divides $\prod_i L_i^{\alpha_i - \beta_i}$, which divides D_H. Thus multiplying every entry of the scaled matrix by D_H gives a matrix over $\mathbb{Z}$ with nonzero determinant. A nonzero integer has modulus at least one, so

$$\begin{aligned} \log |\Delta_H| \geq -M \log D_H &- \sum_{\text{columns } \alpha} \sum_i \alpha_i \log q_i \\ &+ \sum_{\text{rows } \beta} \sum_i \beta_i \log q_i. \end{aligned} \tag{3.8}$$

For each selected column, $\sum_i \alpha_i \log q_i \leq w\alpha \leq H$. Also $0 \leq w_i - \log q_i < 1$, and hence

$$\sum_{\text{rows } \beta} \sum_i \beta_i \log q_i \geq MH\bar{b} - \sum_{\text{rows } \beta} \sum_i \beta_i \geq MH\bar{b} - \frac{MH\theta}{w_*}.$$

For completeness, $\log \operatorname{lcm}(1, \dots, n) \leq \Lambda n$. To see this, the factorial valuation formula shows that each prime power in $(\ell, 2\ell]$ contributes its logarithmic prime weight to $\log \binom{2\ell}{\ell}$; all the other contributions to the latter are nonnegative. Thus the increase of the log least common multiple across this interval is at most $2\ell \log 2$. Summing over dyadic intervals up to the smallest power of two above n gives at most $4n \log 2$. Therefore

$$\frac{\log D_H}{H} \leq \Lambda \sum_i \frac{T_i}{w_i} \leq \frac{\Lambda F_0 m}{v_0} + \Lambda \sum_i \frac{1}{w_i}.$$

Substitution in (3.8) proves (3.7). □

3.2 Translation to entire functions

For every multi-index $a \in \mathbb{Z}^m_{\geq 0}$ and every selected column P, define the entire function

$$f_{a,P}(z) = [u^a] P(e^z, z + u_1, \dots, z + u_m). \tag{3.9}$$

We regard $(f_{a,P})_P$ as a row vector of entire functions, indexed by the selected columns. For $P = Y^h X^\alpha$ it has the explicit entry

$$f_{a,P}(z) = \begin{cases} \binom{\alpha}{a} e^{hz} z^{|\alpha| - |a|}, & a \leq \alpha, \\ 0, & \text{otherwise.} \end{cases} \tag{3.10}$$

In particular only indices a with $wa \leq H$ can contribute.

Lemma 3.2 (Exact row translation)**.** *Every row $\rho = (j, s, \beta)$ of (3.3) is a sum of at most*

$$Q_H = (\lfloor H \rfloor + 1)^{2m}(\lfloor H/v_0 \rfloor + 1) \tag{3.11}$$

scalar multiples of row vectors of the form

$$\left([t^\ell] f_{a,P}(j\omega + \log(1+t))\right)_P, \qquad a \geq \beta, \quad wa \leq H, \quad 0 \leq \ell \leq s.$$

Each scalar ξ occurring with index a satisfies

$$\begin{aligned} |\xi| &\leq \exp\{-\nu w(a-\beta) + HE_{\mathrm{tr}}\}, \\ E_{\mathrm{tr}} &= \frac{\nu}{F_0} + \frac{\log 2}{v_0} + \frac{\log 4 + \log(2K) + \nu}{w_*}. \end{aligned} \tag{3.12}$$

Proof. Put

$$\epsilon_i = j(r_i - \omega), \qquad \tau_i(t) = G_i(t) - \log(1+t).$$

Since $e^{j\omega} = 2^j$, the substitutions in (3.3) can be written as $Y = e^z$, $X_i = z + u_i + \epsilon_i + \tau_i(t)$, where $z = j\omega + \log(1+t)$. Expanding first in the variables u_i, then in the constants ϵ_i and tails τ_i, gives the exact row vector identity

$$\begin{aligned} \mathrm{row}_{j,s,\beta} = \sum_{\substack{a \geq \beta \\ wa \leq H}} \binom{a}{\beta} \sum_{0 \leq d \leq a-\beta} \binom{a-\beta}{d} \epsilon^{a-\beta-d} \sum_{k=0}^{s} \left([t^k] \prod_i \tau_i(t)^{d_i} \right) \\ \cdot \left([t^{s-k}] f_{a,P}(j\omega + \log(1+t))\right)_P. \end{aligned} \tag{3.13}$$

All scalar factors in this identity are independent of the column P. This is therefore an identity to which determinant multilinearity applies directly.

The product of tails vanishes to order at least $\sum_i d_i T_i$. Consequently a nonzero coefficient in (3.13) satisfies

$$\sum_i d_i w_i \leq \frac{v_0}{F_0} \sum_i d_i T_i \leq \frac{v_0 s}{F_0} < \frac{H}{F_0}. \tag{3.14}$$

On $|t| = 1/2$,

$$|\tau_i(t)| \leq \sum_{k \geq T_i} \frac{2^{-k}}{k} \leq 1.$$

Cauchy's coefficient estimate thus bounds the tail coefficient in (3.13) by $2^k \leq 2^s$. From (3.1),

$$|\epsilon_i| \leq 2Kq_i^{-\nu} \leq 2Ke^{\nu} e^{-\nu w_i}.$$

Moreover $\binom{a}{\beta}\binom{a-\beta}{d} \leq 4^{|a|}$ and $|a| \leq H/w_*$. Combining these inequalities and using (3.14) gives

$$\log|\xi| \leq -\nu w(a-\beta) + \frac{\nu H}{F_0} + \frac{H \log 2}{v_0} + \frac{H(\log 4 + \log(2K) + \nu)}{w_*}$$

for each nonzero scalar. A zero scalar obeys the required bound as well. Finally, each coordinate of a and d is at most H because $w_i \geq 1$, while $0 \leq k \leq H/v_0$. This proves the term count and the Lemma. □

3.3 Repeated Taylor degrees force a quadratic saving

The Taylor expansion of interpolation determinants eliminates repeated degrees [9, Theorem 1]. Our rows fall into different transverse-index groups, so this cancellation is used only inside a fixed group. The next bound is uniform over all row choices.

Lemma 3.3 (Collision estimate)**.** *Choose one translated test row from each row expansion in Lemma 3.2, and remove its scalar factor. Let T be the resulting square matrix, and let n_a be the number of its rows with index a. With*

$$c = \frac{\log 2}{4}, \qquad E_{\mathrm{hol}} = \frac{100K}{w_0} + \frac{\log 2}{v_0} + \frac{\log(200K)}{w_*}, \tag{3.15}$$

there is a quantity $\rho_H \to 0$, independent of all these row choices and of the selected columns, such that

$$|\det T| \le \exp\left\{-c\sum_a n_a^2 + MH(E_{\mathrm{hol}} + \rho_H)\right\}. \tag{3.16}$$

Proof. Put $R = 100K$. Equation (3.10) gives, on $|z| \le R$,

$$|f_{a,P}(z)| \le \exp\left\{H\left(\frac{R}{w_0} + \frac{\log(2R)}{w_*}\right)\right\} =: \mathcal{D}_H.$$

Expand

$$f_{a,P}(z) = \sum_{d\ge 0} c_{a,d,P}(z/R)^d.$$

Cauchy's estimate gives $|c_{a,d,P}| \le \mathcal{D}_H$. A row of T is a test of order $\ell \le s$ at some center j. Since

$$|j\omega + \log(1+t)| \le (K-1)\log 2 + \log 2 < 50K = R/2 \qquad (|t| = 1/2),$$

its value on $(z/R)^d$ has modulus at most $2^\ell 2^{-d}$.

Expand the determinant in these power-series rows. The expansion is absolutely convergent: a coefficient determinant has modulus at most $M!\mathcal{D}_H^M$, and the sum of the product bounds 2^{-d} over all degrees is finite. If two rows have the same pair (a,d), their coefficient vectors $(c_{a,d,P})_P$ agree, and that summand vanishes. Thus every nonzero summand has distinct degrees within each group a, and so

$$\sum_{\text{rows}} d \ge \sum_a \binom{n_a}{2}.$$

Use half of the geometric decay for this lower bound and sum the other half freely over all nonnegative degrees. Since $\sum \ell \le MH/v_0$, this gives

$$|\det T| \le M!\mathcal{D}_H^M 2^{MH/v_0} 2^{-\frac{1}{2}\sum_a \binom{n_a}{2}} (1 - 2^{-1/2})^{-M}.$$

Because $\sum_a n_a = M$, the asserted inequality follows with, for example,

$$\rho_H = \frac{\log M + \frac{1}{4}\log 2 - \log(1 - 2^{-1/2})}{H}.$$

Equation (3.5) shows that $\rho_H \to 0$ with the fixed parameters stipulated above. □

3.4 A comparison with two alternatives

Put

$$E_{\mathrm{an}} = E_{\mathrm{tr}} + E_{\mathrm{hol}}. \tag{3.17}$$

Let $0 < A < 1$ set a cutoff AH for transverse weights, and let $0 < \eta < 1$ be a fraction of rows. The main arithmetic cost is $1 - \bar{b}$. In each term of the determinant expansion, if at least ηM rows have $wa \le AH$, they give a collision saving; otherwise the rows with $wa > AH$ supply approximation-error factors. We seek a saving larger than $1 - \bar{b} + E_{\mathrm{ar}} + E_{\mathrm{an}}$, and choose A and η in Section 4.

Proposition 3.4 (Determinant comparison)**.** *Let $0 < A < 1$ and $0 < \eta < 1$, and define*

$$g = \nu(A(1-\eta) - \theta) - (1-\theta), \qquad L = \frac{\eta^2 K \theta^m}{(m+1)v_0 A^m}. \tag{3.18}$$

The approximations (3.1) and the interpolation hypotheses cannot hold simultaneously if

$$g > 0, \qquad E_{\mathrm{ar}} + E_{\mathrm{an}} < g, \qquad cL > 1 + E_{\mathrm{ar}} + E_{\mathrm{an}}. \tag{3.19}$$

Proof. Let

$$D_A(H) = \#\{a \in \mathbb{Z}_{\ge 0}^m : wa \le AH\}, \qquad L_H = \frac{\eta^2 M}{H D_A(H)}.$$

The simplex count, now in dimension m, and (3.5) show that

$$D_A(H) \sim \frac{(AH)^m}{m! \prod_i w_i}, \qquad L_H \longrightarrow L.$$

Expand Δ_H by the row identities in Lemma 3.2. For a choice of terms with indices a_ρ, the product of their scalar factors has modulus at most

$$\exp\left\{-\nu \sum_\rho w a_\rho + \nu M H \bar{b} + M H E_{\mathrm{tr}}\right\}.$$

There are at most Q_H^M term choices. Apply Lemma 3.3 to each one.

Suppose first that at least ηM of its indices obey $wa \le AH$. There are at most $D_A(H)$ such indices, so Cauchy–Schwarz gives

$$\sum_a n_a^2 \ge \frac{(\eta M)^2}{D_A(H)} = MHL_H.$$

The scalar main exponent $-\nu \sum_\rho w(a_\rho - \beta_\rho)$ is nonpositive, because $a_\rho \ge \beta_\rho$. Discarding it leaves an upper bound $-cL_H + E_{\mathrm{an}} + \rho_H$ after division by MH.

In the other case, more than $(1-\eta)M$ indices have $wa > AH$. Hence $\sum_\rho wa_\rho \ge MHA(1-\eta)$. Discarding the nonpositive collision term leaves the upper bound $-\nu(A(1-\eta) - \bar{b}) + E_{\mathrm{an}} + \rho_H$. Taking absolute values before summing the determinant expansion gives

$$\begin{aligned} \frac{\log|\Delta_H|}{MH} &\le E_{\mathrm{an}} + \varepsilon_H + \max\{-cL_H, -\nu(A(1-\eta) - \bar{b})\}, \\ \varepsilon_H &= \rho_H + \frac{\log Q_H}{H} \longrightarrow 0. \end{aligned} \tag{3.20}$$

All bounds here are uniform over the row-term choices.

For the second alternative, the gap between the two main exponents is at least g, since $\bar{b} \le \theta$ gives

$$\nu(A(1-\eta)-\bar{b})-(1-\bar{b}) = \nu A(1-\eta)-1-(\nu-1)\bar{b} \ge g.$$

Under (3.19), both quantities inside the maximum in (3.20), after adding $E_{\mathrm{an}}+\varepsilon_H$, are strictly less than $-(1-\bar{b})-E_{\mathrm{ar}}$ for all sufficiently large H. For the first quantity, use $L_H \to L$ and $1-\bar{b} \le 1$; for the second, use the displayed uniform gap. This contradicts Lemma 3.1. □

4 Choice of parameters and proof of Theorem 1.1

We now combine the interpolation theorem and the determinant estimates. The order of choices is essential: the dimension is fixed before the approximation scales, and all these data are fixed before the polynomial degree tends to infinity.

Lemma 4.1. *For every real $\nu > 2$, there are positive rational numbers θ, A, B, C such that*

$$0 < \theta < A < B < 1, \qquad \nu(A-\theta) > 1-\theta, \qquad C > 1, \quad B < 1/C, \quad B < C\theta < 1. \tag{4.1}$$

One can then choose $0 < \eta < 1$ such that

$$g := \nu\big(A(1-\eta)-\theta\big)-(1-\theta) > 0. \tag{4.2}$$

Proof. Choose a rational b with $1/2 < b < 1-1/\nu$. For a sufficiently small positive rational δ, put $\theta = 1-\delta$ and $A = 1-b\delta$. Then $0 < \theta < A < 1$, and

$$\nu(A-\theta)-(1-\theta) = \big(\nu(1-b)-1\big)\delta > 0,$$

whereas

$$\theta - A^2 = (2b-1)\delta - b^2\delta^2 > 0.$$

It follows that the interval $(A/\theta, 1/A)$ is nonempty. Choose a rational C in this interval. Since $A > \theta$, we have $C > 1$; since $\theta < A$, we also have $C\theta < \theta/A < 1$. Now choose a rational B between A and $\min(1/C, C\theta)$. This proves (4.1). Finally, the strict inequality at $\eta = 0$ allows a sufficiently small $\eta > 0$ in (4.2). □

The two inequalities used in this construction have different roles. The inequality $\nu(A-\theta) > 1-\theta$ supplies the approximation-error gap. The inequality $A^2 < \theta$ permits $A/\theta < C < 1/A$, and hence a choice of B for which $CB < 1$, $C\theta/B > 1$, and $B/A > 1$. With the full-simplex counts of Section 3, these three ratios make the dimension-dependent errors tend to zero while the collision saving tends to infinity. We now choose one finite dimension realizing both requirements; it will remain fixed throughout the degree limit.

Proof of Theorem 1.1. Fix $\nu > 2$. Suppose for a contradiction that there are arbitrarily large positive integers q for which some $p \in \mathbb{Z}$ satisfies

$$\left|\log 2 - \frac{p}{q}\right| \le q^{-\nu}. \tag{4.3}$$

Fix θ, A, B, C, η, g from Lemma 4.1, and put $\varepsilon_0 = \min(g/2, 1/2)$. First choose $F_0 > 2/\theta$ so large that $\nu/F_0 < \varepsilon_0/3$.

For an integer m, set

$$K = \lfloor C^m \rfloor, \qquad w_0 = B^{-m}, \qquad v_0 = 2K\theta^m w_0. \tag{4.4}$$

These are admissible rational interpolation data:

$$K\theta^m \le (C\theta)^m < 1, \qquad K\frac{w_0}{v_0}\theta^m = \frac{1}{2}. \tag{4.5}$$

The relevant limits as $m \to \infty$ are

$$\frac{K}{w_0} \le (CB)^m \longrightarrow 0, \qquad v_0 \sim 2\left(\frac{C\theta}{B}\right)^m \longrightarrow \infty, \qquad \frac{m}{v_0} \longrightarrow 0. \tag{4.6}$$

With the notation of Proposition 3.4, these choices give

$$L = \frac{\eta^2 K\theta^m}{(m+1)v_0 A^m} = \frac{\eta^2 (B/A)^m}{2(m+1)} \longrightarrow \infty. \tag{4.7}$$

Thus, with $\Lambda = 4\log 2$ and $c = (\log 2)/4$, choose m large enough that

$$\frac{\Lambda F_0 m + 2\log 2}{v_0} + \frac{100K}{w_0} < \frac{\varepsilon_0}{3}, \qquad cL > 2. \tag{4.8}$$

Fix this m, and hence K, w_0, v_0.

The remaining constants are now fixed. Choose approximations p_i/q_i satisfying (4.3), with $w_i = \lceil \log q_i \rceil$, successively sufficiently large for Theorem 2.1 and, in addition, for

$$\Lambda \sum_{i=1}^{m} \frac{1}{w_i} + \frac{\theta + \log 4 + \log(2K) + \nu + \log(200K)}{w_*} < \frac{\varepsilon_0}{3}, \qquad w_* = \min_{i>0} w_i. \tag{4.9}$$

This is possible: first impose a sufficiently large lower bound on every w_i, using $\sum_i 1/w_i \le m/w_*$, and then impose the successive geometric thresholds. The assumed denominators are unbounded, so every finite sequence of such requirements can be met. The thresholds for weight separation are independent of the centers. Taking each q_i sufficiently large ensures $p_i \neq 0$, since $q_i^{-\nu} < \log 2$. The centers are $y_j = 2^j$, $c_{ji} = jr_i$, and $r_i = p_i/q_i$. The y_j are nonzero and pairwise distinct, as required by the modified interpolation theorem. The thresholds are independent of both the y_j and c_{ji}.

Using (3.6), (3.12), (3.15), and (3.17), the sum of errors is exactly

$$\begin{aligned} E_{\mathrm{ar}} + E_{\mathrm{an}} = {} & \frac{\nu}{F_0} + \frac{\Lambda F_0 m + 2\log 2}{v_0} + \frac{100K}{w_0} \\ & + \Lambda \sum_i \frac{1}{w_i} + \frac{\theta + \log 4 + \log(2K) + \nu + \log(200K)}{w_*}. \end{aligned}$$

Our three choices make this less than ε_0. In particular it is less than g, while $cL > 2 > 1 + E_{\mathrm{ar}} + E_{\mathrm{an}}$. All hypotheses of Proposition 3.4 now hold, a contradiction. The degree H in that Proposition tends to infinity only after the dimension, weights, and centers just chosen are fixed.

We have excluded (4.3) for arbitrarily large q, for each fixed $\nu > 2$. This gives the required eventual lower bound, in fact with a strict inequality. The contradiction allowed unreduced fractions and zero error. If $\log 2$ were rational, exact representations with arbitrarily large denominators would contradict it; hence $\log 2$ is irrational.

Finally divide the unit interval into N equal parts and apply pigeonhole to $0, \log 2, \dots, N \log 2$ modulo one. There exist $p \in \mathbb{Z}$ and $1 \le q \le N$ such that

$$0 < \left| \log 2 - \frac{p}{q} \right| \le \frac{1}{qN} \le \frac{1}{q^2}.$$

After reduction, the same error is at most the inverse square of the reduced denominator. These errors tend to zero as $N \to \infty$; irrationality forces the reduced denominators to be unbounded. Every exponent less than two therefore occurs infinitely often in the defining inequality. Hence $\mu(\log 2) \ge 2$, and the upper bound already proved gives $\mu(\log 2) = 2$. □

5 Extension to positive rational logarithms

This section belongs to the candidate derivation. Its principal conclusion and its corollaries are conditional on Theorem 2.1, including the assertion that its successive weight thresholds are independent of the interpolation centers. The following argument records all changes needed for a positive rational base; it does not certify the proposed geometric interpolation theorem or claim a new unconditional result.

Theorem 5.1 (Conditional rational-logarithm extension; candidate)**.** *Assume Theorem 2.1 with its stated quantifiers. For every fixed $a \in \mathbb{Q}_{>0} \setminus \{1\}$ and every real $\nu > 2$, there is an integer $Q(a, \nu)$ such that*

$$\left| \log a - \frac{p}{q} \right| > q^{-\nu} \qquad (p \in \mathbb{Z},\ q \in \mathbb{Z},\ q \ge Q(a, \nu)). \tag{5.1}$$

Here fractions need not be reduced, and log *is the real natural logarithm. In particular, conditionally on that interpolation statement, $\log a$ is irrational and $\mu(\log a) = 2$.*

We first give the determinant estimates for this fixed base. Write $a = u/v$ in lowest terms, with positive integers u, v, and put $\omega = \log a$. Fix interpolation data $m, K, w_0, v_0, \theta, w_1, \dots, w_m$ satisfying the hypotheses of Theorem 2.1. Suppose that

$$\left| \omega - \frac{p_i}{q_i} \right| \le q_i^{-\nu}, \qquad q_i \ge 2, \qquad w_i = \lceil \log q_i \rceil, \qquad r_i = p_i/q_i \quad (1 \le i \le m). \tag{5.2}$$

Set $w\alpha = \sum_i w_i \alpha_i$ and $w_* = \min_i w_i$. The centers $y_j = a^j$, $c_{ji} = j r_i$, $0 \le j < K$, have nonzero, pairwise distinct y_j, since $a > 0$ and $a \ne 1$. This is true when $a < 1$ as well.

Choose $F_0 > 2/\theta$ and define

$$T_i = \lceil F_0 w_i / v_0 \rceil, \qquad G_i(t) = \sum_{1 \le k < T_i} \frac{(-1)^{k+1} t^k}{k}, \qquad L_i = \operatorname{lcm}(1, \dots, T_i - 1), \qquad D_H = \prod_i L_i^{\lfloor H / w_i \rfloor}. \tag{5.3}$$

An empty sum is zero, and $L_i = 1$ when $T_i = 1$. Since $v_0 T_i \ge F_0 w_i > w_i/\theta$, the substitution $U_i \mapsto U_i + G_i(t) - \log(1+t)$ and its inverse preserve the V-weighted filtration, where $V = (v_0, w_1/\theta, \dots, w_m/\theta)$. It therefore induces an automorphism on every coefficient packet of weight less than H. Interpolation remains surjective after replacing each $\log(1+t)$ by $G_i(t)$.

Use columns $P = Y^h X^\alpha$ with $w_0 h + w\alpha \le H$, and rows $\rho = (j, s, \beta)$ with $0 \le j < K$ and $v_0 s + w\beta/\theta < H$. The entry is

$$A_{\rho;h,\alpha} = [t^s U^\beta] P\big(a^j (1+t), (j r_i + G_i(t) + U_i)_{i=1}^m\big). \tag{5.4}$$

For all sufficiently large, sufficiently divisible H, interpolation supplies a nonzero square minor Δ_H using every row. Let $M = M_H$ be its size and put

$$\bar{b} = \frac{1}{MH} \sum_{\rho=(j,s,\beta)} w\beta, \qquad M \sim \frac{K\theta^m H^{m+1}}{(m+1)! v_0 \prod_i w_i}, \qquad 0 \le \bar{b} \le \theta. \tag{5.5}$$

The count follows by the simplex-volume argument used in (3.5); the strict row inequality has the same leading term. Every limit in H below keeps all the other data fixed.

5.1 Exact arithmetic clearing for a rational base

Lemma 5.2 (Rational-base arithmetic estimate)**.** *If $w_0 \ge (K-1)\log v$, then*

$$\frac{\log|\Delta_H|}{MH} \ge -(1-\bar{b}) - E_{\mathrm{ar}}, \qquad E_{\mathrm{ar}} = \frac{\Lambda F_0 m}{v_0} + \Lambda \sum_i \frac{1}{w_i} + \frac{\theta}{w_*}, \qquad \Lambda = 4\log 2. \tag{5.6}$$

Proof. Scale the column $Y^h X^\alpha$ by $v^{(K-1)h} \prod_i q_i^{\alpha_i}$ and the row (j,s,β) by $\prod_i q_i^{-\beta_i}$. Its scaled entry is zero unless $\alpha \ge \beta$ componentwise; otherwise it is exactly

$$u^{jh} v^{(K-1-j)h} \prod_i \binom{\alpha_i}{\beta_i} [t^s](1+t)^h \prod_i (jp_i + q_i G_i(t))^{\alpha_i - \beta_i}. \tag{5.7}$$

Both exponents on u and v are nonnegative. The denominator of this entry divides $\prod_i L_i^{\alpha_i-\beta_i}$, which divides D_H. Multiplying every scaled entry by D_H gives an integer matrix with nonzero determinant. Hence

$$\begin{aligned} \log|\Delta_H| \ge & -M\log D_H - \sum_{\text{selected columns } (h,\alpha)} \left((K-1)h\log v + \sum_i \alpha_i \log q_i \right) \\ & + \sum_{\text{rows } (j,s,\beta)} \sum_i \beta_i \log q_i. \end{aligned} \tag{5.8}$$

This is the exact clearing inequality; the numerator u requires no additional factor. For every selected column, $\log q_i \le w_i$ and the hypothesis on w_0 give

$$(K-1)h\log v + \sum_i \alpha_i \log q_i \le w_0 h + w\alpha \le H.$$

Since $0 \le w_i - \log q_i < 1$ and $w\beta < \theta H$, the row contribution is at least

$$MH\bar{b} - \sum_{\text{rows}} |\beta| \ge MH\bar{b} - MH\theta/w_*.$$

The elementary least-common-multiple bound $\log \operatorname{lcm}(1,\dots,n) \le \Lambda n$, established in the proof of Lemma 3.1, gives

$$\frac{\log D_H}{H} \le \Lambda \sum_i \frac{T_i}{w_i} \le \frac{\Lambda F_0 m}{v_0} + \Lambda \sum_i \frac{1}{w_i}.$$

Substitution in (5.8) proves (5.6). □

5.2 Translation and analytic collision with either sign

For a transverse multi-index γ, let

$$f_{\gamma,P}(z) = [U^\gamma]P(e^z, z+U_1, \dots, z+U_m).$$

These entire functions have entries $\binom{\alpha}{\gamma}e^{hz}z^{|\alpha|-|\gamma|}$ when $\gamma \le \alpha$, and zero otherwise. Put $\epsilon_i = j(r_i-\omega)$, $\tau_i = G_i - \log(1+t)$ and $z = j\omega + \log(1+t)$. Because $e^{j\omega} = a^j$, direct expansion gives the row-vector identity

$$\begin{aligned}\mathrm{row}_{j,s,\beta} = \sum_{\substack{\gamma\ge\beta\\ w\gamma\le H}} \binom{\gamma}{\beta} \sum_{0\le d\le\gamma-\beta} \binom{\gamma-\beta}{d} \epsilon^{\gamma-\beta-d} \sum_{k=0}^{s} \left([t^k] \prod_i \tau_i(t)^{d_i} \right) \\ \cdot \left([t^{s-k}] f_{\gamma,P}(j\omega + \log(1+t)) \right)_P.\end{aligned} \tag{5.9}$$

All its scalar coefficients are independent of P. The identity has at most

$$Q_H = (\lfloor H \rfloor + 1)^{2m} (\lfloor H/v_0 \rfloor + 1)$$

terms, because $w_i \ge 1$. Each nonzero tail coefficient has $\sum_i d_i T_i \le k \le s$, hence $\sum_i d_i w_i < H/F_0$. On $|t| = 1/2$, $|\tau_i(t)| \le 1$; Cauchy's estimate bounds its product's t^k coefficient by 2^k. Furthermore $|\epsilon_i| \le 2Ke^\nu e^{-\nu w_i}$ and $\binom{\gamma}{\beta}\binom{\gamma-\beta}{d} \le 4^{|\gamma|}$. Consequently every scalar ξ with index γ satisfies

$$|\xi| \le \exp\{-\nu w(\gamma-\beta) + HE_{\mathrm{tr}}\}, \qquad E_{\mathrm{tr}} = \frac{\nu}{F_0} + \frac{\log 2}{v_0} + \frac{\log 4 + \log(2K) + \nu}{w_*}. \tag{5.10}$$

The argument uses absolute values throughout, so it does not require $\omega > 0$ or any sign condition on the p_i.

Set

$$R_a = 4K(1 + |\log a|), \qquad c = \frac{\log 2}{4}, \qquad E_{\mathrm{hol},a} = \frac{R_a}{w_0} + \frac{\log 2}{v_0} + \frac{\log(2R_a)}{w_*}. \tag{5.11}$$

Choose one test row from each row expansion, remove its scalar, and write n_γ for the number of resulting rows with index γ. For the resulting matrix T, we claim

$$|\det T| \le \exp\left\{ -c\sum_\gamma n_\gamma^2 + MH(E_{\mathrm{hol},a} + \rho_H) \right\}, \qquad \rho_H \longrightarrow 0, \tag{5.12}$$

uniformly over its row choices and the selected columns. Indeed, on $|z| \le R_a$ the monomial formula gives

$$|f_{\gamma,P}(z)| \le \exp\left\{ H\left(\frac{R_a}{w_0} + \frac{\log(2R_a)}{w_*} \right) \right\} =: \mathcal{D}_H.$$

Write $f_{\gamma,P}(z) = \sum_{d\ge0} c_{\gamma,d,P}(z/R_a)^d$; then $|c_{\gamma,d,P}| \le \mathcal{D}_H$. The logarithm in the test is its analytic germ on $|t| < 1$. Its power series gives $|\log(1+t)| \le \log 2$ on $|t| = 1/2$, so

$$|j\omega + \log(1+t)| \le (K-1)|\omega| + \log 2 < R_a/2.$$

Thus a test of order $\ell \le s$ on $(z/R_a)^d$ has modulus at most $2^\ell 2^{-d}$. The determinant expansion in these series is absolutely convergent, since each coefficient determinant is bounded by $M!\mathcal{D}_H^M$ and the geometric series converge. Any summand with two equal pairs (γ, d) vanishes, since its two coefficient rows coincide. Every remaining summand therefore has $\sum_{\mathrm{rows}} d \ge \sum_\gamma \binom{n_\gamma}{2}$. Use half of the factor $2^{-\sum d}$ for this saving and sum the other half freely. Since $\sum \ell \le MH/v_0$, this yields

$$|\det T| \le M!\mathcal{D}_H^M 2^{MH/v_0} 2^{-\frac12\sum_\gamma \binom{n_\gamma}{2}} (1 - 2^{-1/2})^{-M}.$$

Equation (5.12) follows, for example with

$$\rho_H = \frac{\log M + \frac{1}{4}\log 2 - \log(1-2^{-1/2})}{H} \longrightarrow 0,$$

where (5.5) gives the limit.

For $0 < A < 1$ and $0 < \eta < 1$, put

$$g = \nu(A(1-\eta)-\theta) - (1-\theta), \qquad L = \frac{\eta^2 K\theta^m}{(m+1)v_0 A^m}, \qquad E_a = E_{\mathrm{ar}} + E_{\mathrm{tr}} + E_{\mathrm{hol},a}. \tag{5.13}$$

The contradiction conditions are exactly

$$w_0 \geq (K-1)\log v, \qquad g > 0, \qquad E_a < g, \qquad cL > 1 + E_a. \tag{5.14}$$

Here is the full comparison. Define $D_A(H) = \#\{\gamma \in \mathbb{Z}^m_{\geq 0} : w\gamma \leq AH\}$ and $L_H = \eta^2 M/(HD_A(H))$. Simplex counts give $D_A(H) \sim (AH)^m/(m!\prod_i w_i)$ and $L_H \to L$. In one term of the multilinear expansion of Δ_H, if at least ηM indices have $w\gamma \leq AH$, then $\sum n_\gamma^2 \geq (\eta M)^2/D_A(H) = MHL_H$. The scalar main exponent $-\nu\sum w(\gamma-\beta)$ is nonpositive, so this term has normalized logarithmic upper bound $-cL_H + E_{\mathrm{tr}} + E_{\mathrm{hol},a} + \rho_H$. Otherwise more than $(1-\eta)M$ indices have $w\gamma > AH$. The scalar estimate then gives normalized main exponent at most $-\nu(A(1-\eta)-\bar{b})$; discard the nonpositive collision term. There are at most Q_H^M expansion terms. Taking absolute values before summing gives

$$\frac{\log|\Delta_H|}{MH} \leq E_{\mathrm{tr}} + E_{\mathrm{hol},a} + \varepsilon_H + \max\{-cL_H, -\nu(A(1-\eta)-\bar{b})\}, \quad \varepsilon_H = \rho_H + \frac{\log Q_H}{H} \to 0. \tag{5.15}$$

For the second alternative the gap over $1-\bar{b}$ is at least g, because

$$\nu(A(1-\eta)-\bar{b}) - (1-\bar{b}) = \nu A(1-\eta) - 1 - (\nu-1)\bar{b} \geq g.$$

For the first alternative use $1-\bar{b} \leq 1$ and $L_H \to L$. Under (5.14), both upper bounds in (5.15) are strictly less than the lower bound $-(1-\bar{b}) - E_{\mathrm{ar}}$ for sufficiently large H. This contradicts (5.6).

5.3 Finite parameter choice for each fixed base

Proof of Theorem 5.1. Fix $a = u/v \in \mathbb{Q}_{>0} \setminus \{1\}$ and $\nu > 2$ before making any dimension or approximation choice. Suppose that arbitrarily large integers q admit $p \in \mathbb{Z}$ with $|\log a - p/q| \leq q^{-\nu}$. Choose θ, A, B, C, η as in Lemma 4.1, so

$$0 < \theta < A < B < 1, \qquad C > 1, \qquad CB < 1, \qquad B < C\theta < 1, \qquad g > 0.$$

Let $\varepsilon_0 = \min(g/2, 1/2)$ and choose $F_0 > 2/\theta$ such that $\nu/F_0 < \varepsilon_0/3$. For a variable positive integer m, set

$$K = \lfloor C^m \rfloor, \qquad w_0 = B^{-m}, \qquad v_0 = 2K\theta^m w_0.$$

The interpolation volume hypotheses hold, since $K\theta^m \leq (C\theta)^m < 1$ and $K(w_0/v_0)\theta^m = 1/2$. Moreover

$$\frac{K}{w_0} \leq (CB)^m \to 0, \qquad v_0 \sim 2(C\theta/B)^m \to \infty, \qquad m/v_0 \to 0, \qquad L = \frac{\eta^2(B/A)^m}{2(m+1)} \to \infty.$$

Because a was already fixed, these limits also give

$$R_a/w_0 = 4(1+|\log a|)K/w_0 \to 0, \qquad (K-1)\log v/w_0 \to 0.$$

Consequently one finite m can be chosen such that

$$w_0 \geq (K-1)\log v, \qquad \frac{\Lambda F_0 m + 2\log 2}{v_0} + \frac{R_a}{w_0} < \varepsilon_0/3, \qquad cL > 2. \tag{5.16}$$

Fix this m, hence K, w_0, v_0, R_a and all the constants in this display.

Choose m approximations as in (5.2) successively, with denominators sufficiently large to satisfy the successive weight thresholds of Theorem 2.1 and

$$\Lambda \sum_i \frac{1}{w_i} + \frac{\theta + \log 4 + \log(2K) + \nu + \log(2R_a)}{w_*} < \varepsilon_0/3. \tag{5.17}$$

To realize the last inequality, first impose a sufficiently large common lower bound on all w_i, using $\sum_i 1/w_i \leq m/w_*$. Then impose the geometric threshold for each weight after its predecessors have been chosen. The assumed good denominators are unbounded, so every one of these finitely many lower bounds is attainable, even if those denominators are sparse. The geometric weight thresholds are independent of the centers subsequently formed from these choices. No nonzero-numerator hypothesis is needed: the distinctness of a^j does not involve the p_i.

The total error is exactly

$$\begin{aligned} E_a = \frac{\nu}{F_0} + \frac{\Lambda F_0 m + 2\log 2}{v_0} + \frac{R_a}{w_0} + \Lambda \sum_i \frac{1}{w_i} \\ + \frac{\theta + \log 4 + \log(2K) + \nu + \log(2R_a)}{w_*} < \varepsilon_0. \end{aligned}$$

Thus $E_a < g$ and $cL > 2 > 1 + E_a$, and all conditions (5.14) hold. After fixing these weights and centers, let H tend to infinity through sufficiently divisible integers above the interpolation threshold for these particular centers. The comparison (5.15)–(5.6) is a contradiction. No uniform threshold in H over different bases or centers has been used.

The good denominators for the fixed a, ν are therefore bounded, which proves (5.1). If ω were rational, its exact representations with arbitrarily large positive denominators would contradict that bound, so ω is irrational. It follows that $\mu(\omega) \leq 2$. For completeness, pigeonhole applied to $0, \omega, \ldots, N\omega$ modulo one gives $p \in \mathbb{Z}$ and $1 \leq q \leq N$ with $0 < |\omega - p/q| \leq 1/(qN) \leq 1/q^2$. The errors tend to zero as $N \to \infty$; after reduction the denominators must be unbounded, since ω is irrational. Every exponent less than two consequently occurs infinitely often, so $\mu(\omega) \geq 2$. This completes the conditional proof. □

5.4 Rational scaling and further conditional consequences

Lemma 5.3 (Invariance under nonzero rational scaling)**.** *If x is an irrational real number and $b \in \mathbb{Q} \setminus \{0\}$, then bx is irrational and $\mu(bx) = \mu(x)$.*

Proof. Write $b = r/s$ with $r \in \mathbb{Z} \setminus \{0\}$, $s \geq 1$ and $\gcd(|r|, s) = 1$. Irrationality of bx follows by division by b. Suppose a positive exponent τ occurs infinitely often for x, with coprime p, q, $q \geq 2$, and $0 < |x - p/q| < q^{-\tau}$. Reduce $rp/(sq)$ to P/Q, and set $d = \gcd(|rp|, sq)$. Since $\gcd(p, q) = 1$, a Bezout identity for p, q shows that d divides rs: multiply that identity by rs, noting that d divides both rp and sq. Thus $1 \leq d \leq |r|s$ and

$$q/|r| \leq Q = sq/d \leq sq.$$

In particular Q is unbounded. Regardless of the sign of r,

$$0 < \left|bx - \frac{P}{Q}\right| = \frac{|r|}{s}\left|x - \frac{p}{q}\right| < \frac{|r|}{s}q^{-\tau} \le |r|s^{\tau-1}Q^{-\tau}.$$

For every $0 < \lambda < \tau$, the fixed constant $|r|s^{\tau-1}$ is less than $Q^{\tau-\lambda}$ for all sufficiently large Q. Hence λ occurs infinitely often for bx. Taking suprema and then letting λ increase to each such τ gives $\mu(bx) \ge \mu(x)$, also when that supremum is infinite. Apply the same argument to $1/b$ and bx to obtain the reverse inequality. □

Corollary 5.4 (Conditional consequences; candidate)**.** *Assume Theorem 2.1 with its stated quantifiers. Every nonzero finite rational linear combination*

$$\omega = \sum_{k=1}^{n} b_k \log a_k, \qquad b_k \in \mathbb{Q}, \quad a_k \in \mathbb{Q}_{>0},$$

is irrational and has $\mu(\omega) = 2$. In particular this applies to $\log a$ whenever $a > 0$ is algebraic, $a \neq 1$, and $a^N \in \mathbb{Q}$ for some integer $N \ge 1$.

Proof. Choose a positive integer N such that all Nb_k are integers. The real logarithm identities give

$$N\omega = \log r, \qquad r = \prod_{k=1}^{n} a_k^{Nb_k} \in \mathbb{Q}_{>0}.$$

Negative integer exponents preserve positive rationality. Since $\omega \neq 0$, one has $r \neq 1$. Theorem 5.1 gives irrationality and exponent two for $\log r$, conditionally on the interpolation hypothesis. Lemma 5.3 with the factor $1/N$ gives the assertion for ω. For the final statement, $r = a^N$ is positive rational and is not one, and $\log a = (1/N)\log r$; apply the same argument. □

The conditional statements above add no interpolation dependency beyond Theorem 2.1. They assert a base-dependent eventual denominator threshold, rather than a common threshold over all rational bases. They make no claim about logarithms of arbitrary positive algebraic numbers whose positive integer powers are never rational.

6 Audit interface for the candidate proof

The key dependency is Theorem 2.1, including its center-independent successive weight thresholds. Its proposed proof passes through the local transverse multiplicity bound, persistent derivative-ideal components, the residue obstruction forcing constant Y, and the ordinary blow-up and Serre-vanishing interpolation argument. Replacing $Y = 1$ by $Y = 2^j$ is a new mathematical step.

If that geometric statement and the determinant estimates hold as written, Section 4 selects a finite dimension for each fixed $\nu > 2$, then chooses approximation denominators and their separated weights, then fixes the centers, and only finally lets H tend to infinity. Excluding unbounded ν-good approximations for every $\nu > 2$ gives $\mu(\log 2) \le 2$; the final Dirichlet argument gives the reverse bound. The endpoint equality is therefore a conditional consequence of the complete proposed chain, not a deduction from finite numerical tests.

Internal readers have not located a definite invalid inference in the inspected chain. That observation is not proof certification. The new interpolation theorem, its hypotheses, and its use in the determinant contradiction should be checked before citing this document as a result.

References

[1] OpenAI, *The irrationality exponent of π is* 2, 24 September 2026. Manuscript and source repository.

[2] R. Marcovecchio, *The Rhin–Viola method for log 2*, Acta Arith. 139 (2009), 147–184. Publisher PDF.

[3] P. Mondal, *How many zeroes?*, Draft III, arXiv:1806.05346v5, Corollary VIII.3, for isolated local intersection multiplicities.

[4] The Stacks Project, 02OS, 02ND, 02NS, 0AG6, 0AG7, 0B5U, 0BXX.

[5] R. Lazarsfeld, *Positivity in Algebraic Geometry I*, Springer, 2004, Corollary 1.4.10.

[6] J.-P. Demailly, *Singular hermitian metrics on positive line bundles*, LNM 1507 (1992), 87–104, Section 6.

[7] P. Philippon, *Lemmes de zéros dans les groupes algébriques commutatifs*, Bull. SMF 114 (1986), 355–383; errata 115 (1987).

[8] M. Laurent, *Sur quelques résultats récents de transcendance*, Astérisque 198–200 (1991), 209–230, Section 6.

[9] M. Laurent, *Interpolation determinants of exponential polynomials*, Publ. Math. Debrecen 56 (2000), 457–473, Theorem 1.